

① $A_o = -100 \text{ V}$

$A_{o1} \rightarrow \text{inverting}$

$A_{o2} \rightarrow \text{non-inverting}$

$$a(jf) = \frac{5 \times 10^5}{1 + jf/f_o}$$

$$a_o f_o = f_c = 15 \text{ MHz} \rightarrow f_o = 30 \text{ Hz}$$

a) $f_{B1} = f_{B2} \rightarrow \beta_1 = \beta_2 = \beta \quad (f_c = f_o/\beta)$

for subcircuit 1: $A_{o1} = -R_2/R_1$, $\beta_1 = 1/(1 + R_2/R_1) = 1/(1 - A_{o1})$
 $A_{o1} = 1 - 1/\beta_1$

2: $A_{o2} = 1 + R_2/R_1$, $\beta_2 = 1/(1 + R_2/R_1) = 1/A_{o2}$

$$A_o = 100 = A_{o1} A_{o2} = (1 - 1/\beta_1)(1/\beta_2) = (1 - 1/\beta)(1/\beta)$$

$$-100\beta^2 = \beta - 1 \rightarrow \beta = 0.095$$

$$f_{B1} = \beta f_c = 1.425 \text{ MHz}$$

$$A_{o2} = 1/\beta = 10.51$$

$$f_{-3dB} = \sqrt{2^{1/2} - 1} f_o = 918.3 \text{ kHz}$$

$$A_{o1} = -100/10.51 = -9.5$$

b) let $A_{o1} = -A_{o2}$, what are β_1 & β_2 ?

$$A_{o1} A_{o2} = -100 = -A_{o2}^2 \rightarrow A_{o2} = 10, A_{o1} = -10$$

$$\beta_1 = 1/(1 - A_{o1}) = 1/11, \quad \beta_2 = 1/10$$

$$f_{B1} = \frac{1}{11} f_c = 1.36 \text{ MHz}$$

$$f_{B2} = \frac{1}{10} f_c = 1.5 \text{ MHz}$$

full model:

$$A = -100 / ((1 + jf/1.36\text{M})(1 + jf/1.5\text{M}))$$

$$f_{-3dB} \rightarrow \frac{|-100|}{\sqrt{2}} = \frac{100}{\sqrt{1 + f_{-3dB}^2/1.36\text{M}^2} \sqrt{1 + f_{-3dB}^2/1.5\text{M}^2}}$$

$$\rightarrow \sqrt{2} = \sqrt{1 + f_{-3dB}^2/1.36\text{M}^2} \sqrt{1 + f_{-3dB}^2/1.5\text{M}^2}$$

$$2 = (1 + f_{-3dB}^2/1.36\text{M}^2)(1 + f_{-3dB}^2/1.5\text{M}^2)$$

$$f_{-3dB} = 917.7 \text{ kHz}$$

c) $A_{o1} = -50 \text{ V}$

$\rightarrow$

$$A_{o2} = 2 \text{ V}$$

$$\beta_1 = 1/(1 + 50) = 0.0196$$

$$\beta_2 = 1/2 = 0.5$$

$$f_{B1} = 0.0196 \cdot f_c = 294 \text{ kHz}$$

$$f_{B2} = 0.5 f_c = 7.5 \text{ MHz}$$

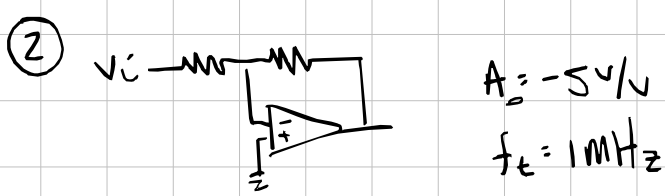
since $f_{B1} \ll f_{B2}$, then $f_{-3dB} \approx f_{B1}$

Show this:

$$\sqrt{2} = \sqrt{1 + f_{-3dB}^2/f_{B1}^2} \sqrt{1 + f_{-3dB}^2/f_{B2}^2}$$

$$2 = (1 + f_{-3dB}^2/f_{B1}^2)(1 + f_{-3dB}^2/f_{B2}^2)$$

$$\rightarrow f_{-3dB} = 293.5 \text{ kHz}$$



from previous problem:

$$\beta = 1/(1 - A_0) = 1/6 \rightarrow f_0 = \frac{1}{6} 1 \text{ MHz} = 167 \text{ kHz}$$

a) full power bandwidth must be $\geq 167 \text{ kHz}$

$$f_{PB} = \frac{SR}{2\pi V_{sat}} \quad V_{sat} = 5 \text{ V}$$

$$167 \text{ k} = SR / 2\pi \cdot 5$$

$$SR \geq 5.25 \times 10^6 \text{ V/s}$$

$$SR \geq 5.25 \text{ V/}\mu\text{s}$$

b) input is $0.625 \sin(2 \times 10^5 t + 75^\circ)$, what is minimum SR?

$$\text{Assume } f_{PB} = 2 \times 10^5 \text{ rad/s} \rightarrow 31.8 \text{ kHz}$$

$$V_{out \text{ max}} = 0.625 \cdot A_0 = 3.125 \text{ V}$$

$$31.8 \text{ kHz} = SR / 2\pi \cdot 3.125 \text{ V}$$

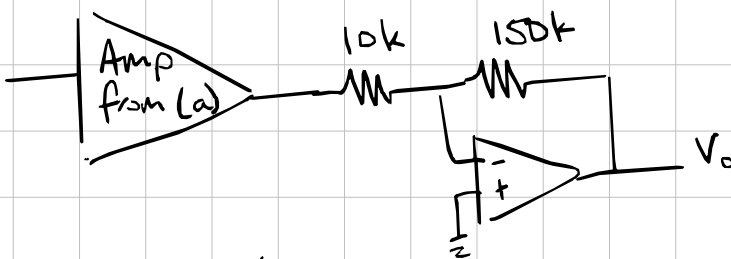
$$SR \geq 625000 \text{ V/s}$$

$$SR \geq 0.625 \text{ V/}\mu\text{s}$$

c) See next page

Note: Plot with 100% SR ($V_0 - SR_{100}$) is not equal to ideal! This is due to AC effects from small signal bandwidth (small gain loss and phase delay)

d) Want $A_0 = 75$, have -5
 $\therefore$ need another inverting amp with gain $75/-5 = -15$



$$\beta = 10\text{k} / (10\text{k} + 150\text{k}) = 1/16 = 0.0625$$

$$f_{B2} = 0.0625 (1 \text{ MHz}) = 62.5 \text{ kHz}$$

knowing $f_{B1} = 167 \text{ kHz}$:

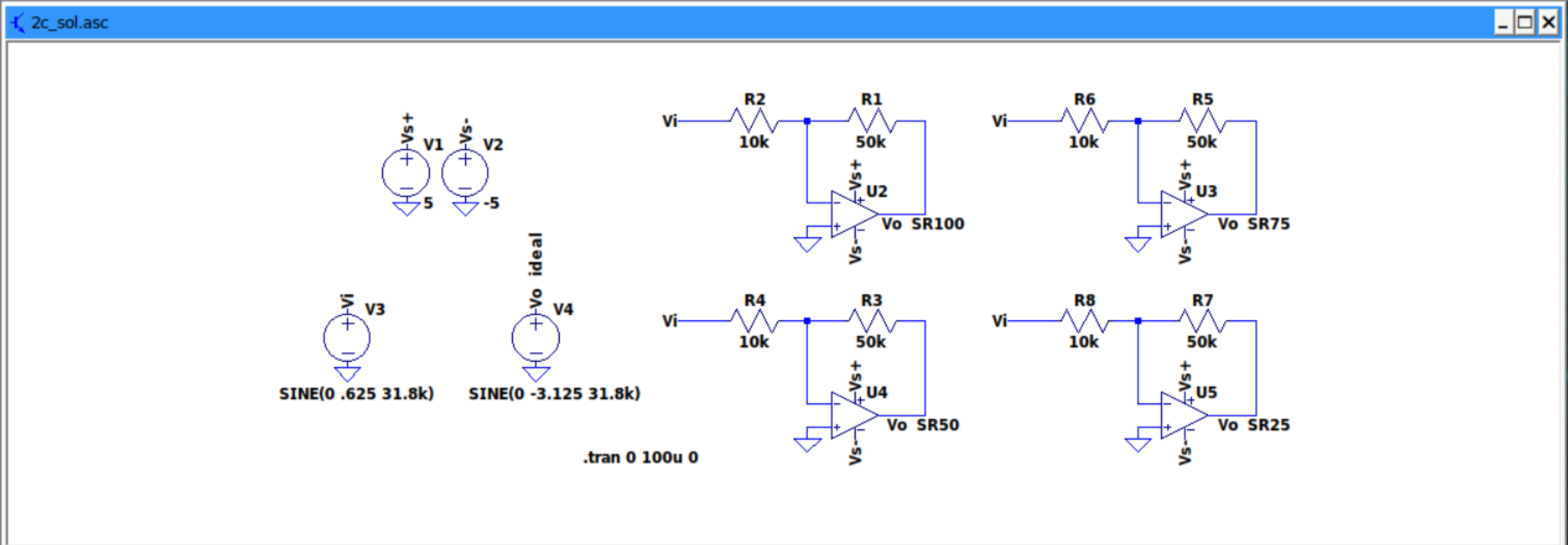
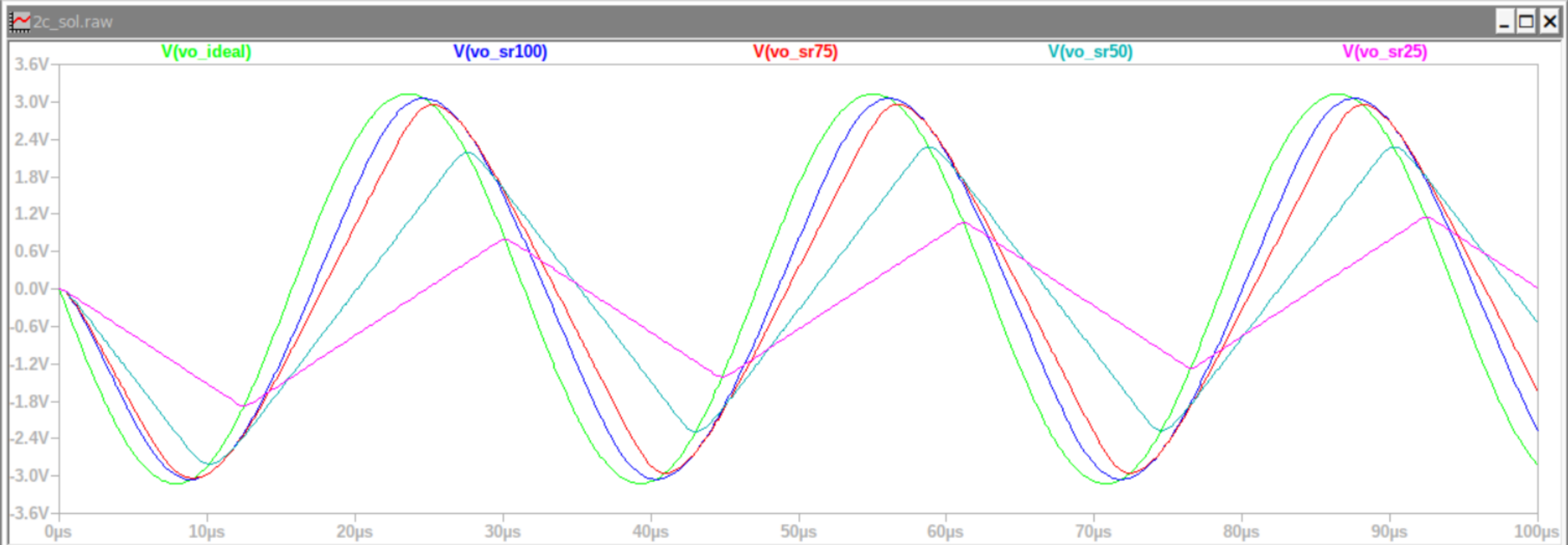
$$2 = \left(1 + \frac{f_{-3dB}^2}{f_{B1}^2}\right) \left(1 + \frac{f_{-3dB}^2}{f_{B2}^2}\right)$$

$$f_{-3dB} = 55.9 \text{ kHz}$$

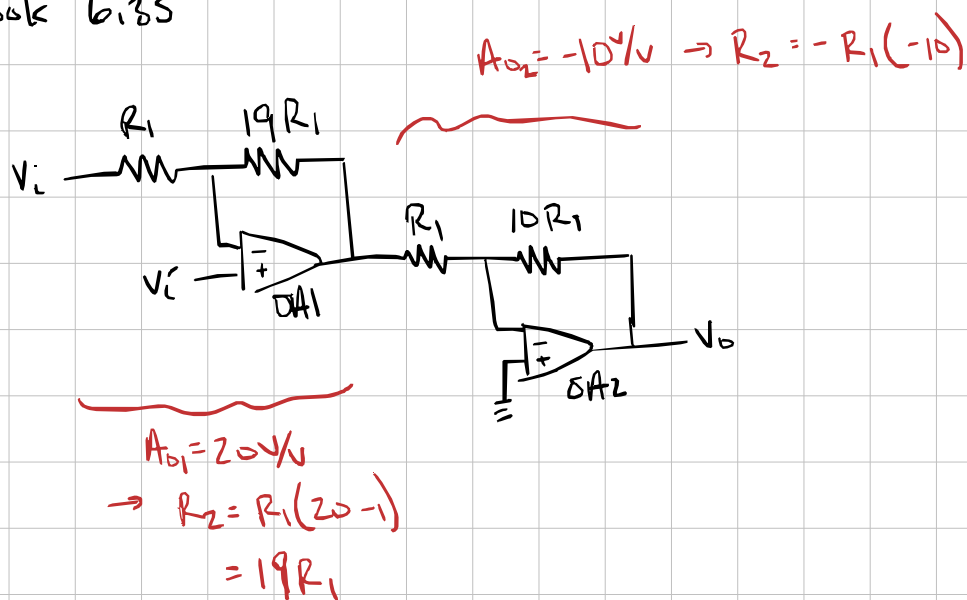
$$f_T = |A_0| f_0 = (75)(55.9 \text{ k})$$

$$f_T = 4.2 \text{ MHz}$$

NOTE: This is the value that the question is asking for BUT it is invalid as a cascaded amplifier does not behave like a GBWP output since decay is -40 dB/dec , not -20 dB/dec
 $\therefore$ The posed question is incorrect.



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want bandwidth to be 100 kHz , assume $f_{B1} = f_{B2}$:

$$f_{-3dB} = 100 \text{ kHz} = \sqrt{2^{1/2}-1} f_{B1}$$

$$\rightarrow f_{B1} = f_{B2} = 155 \text{ kHz}$$

f_t for OA1: we know non-inv. amps have $f_T = f_t$

$$\text{therefore } f_{t1} = f_{T1} = |A_{01}| f_{B1}$$

$$f_{t1} = 3.1 \text{ MHz}$$

f_t for OA2: $f_T \neq f_t$ need β ...

$$\beta = R_1 / (R_1 + 10R_1) = 1/11$$

$$f_B = f_{t2} \beta$$

$$f_{t2} = 1.7 \text{ MHz}$$

Slew Rates: want $V_o = 5 \text{ V}_{rms} \rightarrow 7.07 \text{ V}_{peak}$
at 100 kHz

V_o comes from OA2, therefore:

$$100 \text{ kHz} = \frac{SR_2}{2\pi \cdot 7.07}$$

$$SR_2 = 4.44 \text{ V}/\mu\text{s}$$

V_o of OA1 is V_i of OA2:

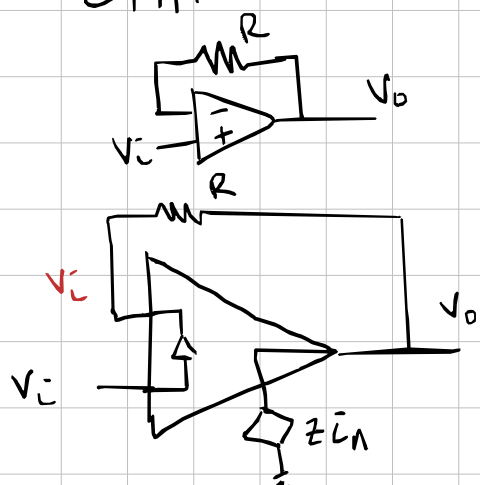
$$V_{o1} = V_{i2} = V_{o2} / |A_{02}| = 0.707 \text{ V}_{peak}$$

$$100 \text{ kHz} = \frac{SR_1}{2\pi \cdot 0.707}$$

$$SR_1 = 0.44 \text{ V}/\mu\text{s}$$

④

CFA:



show $V_o = V_i$

$$V_o = Z i_n \Rightarrow i_n = V_o / Z$$

$$i_n = (V_i - V_o) / R$$

$$\frac{V_o}{Z} = (V_i - V_o) / R$$

$$V_o \left(1 + \frac{R}{Z} \right) = V_i$$

$$V_o = V_i \quad \text{IF} \quad 1 + \frac{R}{Z} = 0$$

we know for a CFA that the feedback resistor must be $R_f \ll Z$,

therefore, $1 + \frac{R}{Z} \approx 1$

and $A_{ideal} = 1 \checkmark$

Error at frequency:

$$Z = R_{eq} \parallel Z_{Ceq} = \frac{R_f / j\omega C_f}{R_f + 1 / j\omega C_f} = \frac{R_f}{1 + jf / f_b}$$

$$R_{eq} = 500 \text{ k}\Omega$$

$$C_{eq} = 2 \text{ pF}$$

$$f_b = 1 / (2\pi R_{eq} C_{eq})$$

$$f_b = 159 \text{ kHz}$$

$$Z = 500 \text{ k} / (1 + jf / 159 \text{ k})$$

with $R = 10 \text{ k}\Omega$

$$A = \frac{V_o}{V_i} = 1 / (1 + (10 \text{ k} / 500 \text{ k}) / (1 + jf / 159 \text{ k}))$$

$$A = 1 / (1 + \frac{1 + jf / 159 \text{ k}}{50}) = 50 / (51 + jf / 159 \text{ k})$$

$$|A| = 50 / \sqrt{51^2 + f^2 / 159 \text{ k}^2}$$

$$GE = \frac{A_{ideal} - |A|}{A_{ideal}} \cdot 100\%$$

f =	0 Hz	100 kHz	1 MHz	10 MHz
A =	0.9804	0.9803	0.9730	0.6175
GE =	1.96%	1.97%	2.70%	38.3%